Physical Reality Interpretation Layer (PRIL™)

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Parent Standard: CLIA® — Cognitive Layered Intelligence Architecture
Compatibility: OOF® Methodology OS™ · **CLIA®** · **SIMULOS®** · **RIS** · **INTEGROS®** · **EVIP®** · ORGS™
AI-Readable: Yes
Authority: OOF®
Protection: MIP® — Methodological Intellectual Property
Canonical Language: English (UCL™)

Canonical Definition System

Physical Reality Interpretation Layer (PRIL™) defines the methodological conditions under which autonomous systems interpret physical-world signals, spatial environments, human presence, movement, proximity, objects, and embodied interaction contexts before generating operational decisions or physical actions.

PRIL™ does not define sensors as hardware components.

PRIL™ defines the interpretation layer through which sensor-derived signals become operational understanding inside autonomous systems.

A. Standard Abstract

Autonomous systems increasingly operate in physical environments where perception errors may produce real-world consequences.

Vision systems, lidar, radar, thermal inputs, spatial mapping tools, and multimodal sensing do not create safe behavior by themselves.

Safety depends on how physical reality is interpreted.

PRIL™ establishes the interpretation layer required for systems that must understand:

- space
- distance
- movement
- human proximity
- object position
- environmental change
- physical interaction risk

before acting in the real world.

B. Core Principle

A physical autonomous system cannot be considered operationally reliable unless perception inputs are interpreted through a governed physical reality interpretation layer before execution.

C. Scope

PRIL™ applies to:

- humanoid robots
- industrial robots
- drones
- autonomous vehicles
- robotic arms
- warehouse automation
- embodied AI systems
- spatial AI systems
- physical-world autonomous agents

PRIL™ may be used wherever AI systems interpret physical reality before movement, interaction, navigation, manipulation, or operational execution.

D. Core Conditions

A PRIL™-compatible system must define:

- how physical signals enter interpretation
- how human presence is detected
- how proximity is evaluated
- how movement is interpreted
- how spatial uncertainty is handled
- how unsafe physical conditions are recognized
- how execution is restricted when interpretation confidence is insufficient

E. Interpretation Integrity Condition

No physical action may proceed when physical reality interpretation is absent, degraded, contradictory, or below the required operational confidence threshold.

If the system cannot interpret the physical environment reliably, execution must be restricted.

Non-bypassable logic applies at validated state only.

F. Methodology

Implementation of PRIL™ requires:

- 1. define the physical environment relevant to execution
- 2. define the perception inputs entering interpretation
- 3. define how spatial, human, and object conditions are recognized
- 4. define how uncertainty, ambiguity, or conflict is handled
- 5. define how unsafe conditions restrict or stop execution
- 6. verify that physical interpretation remains structurally linked to execution logic

A system may be relied upon only after physical reality interpretation conditions are established.

G. Invalid Conditions

A system is considered PRIL™-invalid if:

- physical signals are used without governed interpretation
- human presence is not reliably recognized where required
- proximity or movement conditions remain undefined
- sensor conflict is ignored or silently bypassed
- spatial uncertainty does not restrict execution
- physical action proceeds despite degraded interpretation conditions

A sensor stack alone does not create valid physical interpretation.

H. Cross-Module Dependency

PRIL™ depends on:

- **CLIA®** for interpretation architecture
- **SIMULOS®** for simulated physical-world validation
- **RIS** for runtime enforcement
- **INTEGROS®** for non-bypassable integrity boundaries
- **EVIP®** for ethical human interaction constraints

I. System Position

PRIL™ functions as:

- a derived interpretation layer under **CLIA®**
- a physical-world meaning layer for autonomous systems
- a bridge between sensing input and operational execution
- a governance condition for embodied runtime environments

It does not define hardware sensing itself.

It defines how physical reality becomes operationally interpretable before action.

J. Compatibility Statement

A system may declare:

- **“Physical Reality Interpretation Layer Compatible”**
- only if physical-world perception is interpreted, governed, validated, and operationally restricted under defined conditions

before physical execution.

Canonical Closing Statement

Sensors do not create understanding. Physical reality becomes operationally usable only when perception is interpreted through governed system logic.

Recommended Modules under PRIL™

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1. Human Proximity Interpretation Module

Defines how systems interpret human presence, distance, movement, and unsafe closeness.

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2. Spatial Environment Interpretation Module

Defines how systems interpret space, obstacles, terrain, object location, and environmental structure.

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3. Movement Prediction Interpretation Module

Defines how systems interpret human, robotic, vehicle, or object movement trajectories before action.

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4. Sensor Conflict Resolution Module

Defines how systems handle conflicting input from lidar, camera, radar, thermal, depth, or multimodal systems.

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5. Physical Interaction Safety Module

Defines how systems restrict touch, contact, manipulation, collision risk, and embodied interaction with humans or objects.

Module Architecture

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Canonical Source

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